

"Ayushasthra"

Ayurvedic Medicine Identification and Health Management App

Group Number : R25-068



Introduction

Our Ayurvedic Medicine Identification and Health Management App combines the ancient wisdom of Ayurveda with modern technology.

By leveraging

- > Artificial intelligence.**
- > Image processing**
- > Data analytics**

This app makes

- > Ayurvedic care more accessible and personalized.**
- > Helps users identify medicinal plants**
- > Gain health insights**

By focusing demand for holistic and natural health solutions, this app offers an accurate, user-friendly way to bridge traditional practices and modern needs.

Mainly Focusing on,

- > AI-Powered Tongue Diagnostics for Health Monitoring and Wellness Guidance**
- > AI-powered Identification of Skin Diseases and Analysis of Skin Tone & Hydration levels**
- > Ayurvedic medicinal plant identification with detailed usage guidelines.**
- > Mental Well-being Assessment through Facial Emotion Analysis**

Research Questions

- Ayurvedic diagnosis is limited by practitioner dependency, delaying treatment and reducing accessibility.

Skin diseases lack integrated solutions combining Ayurvedic and modern approaches. Health tools fail to offer personalized care by excluding Ayurvedic data, while difficulties in identifying medicinal plants further restrict remedy access.

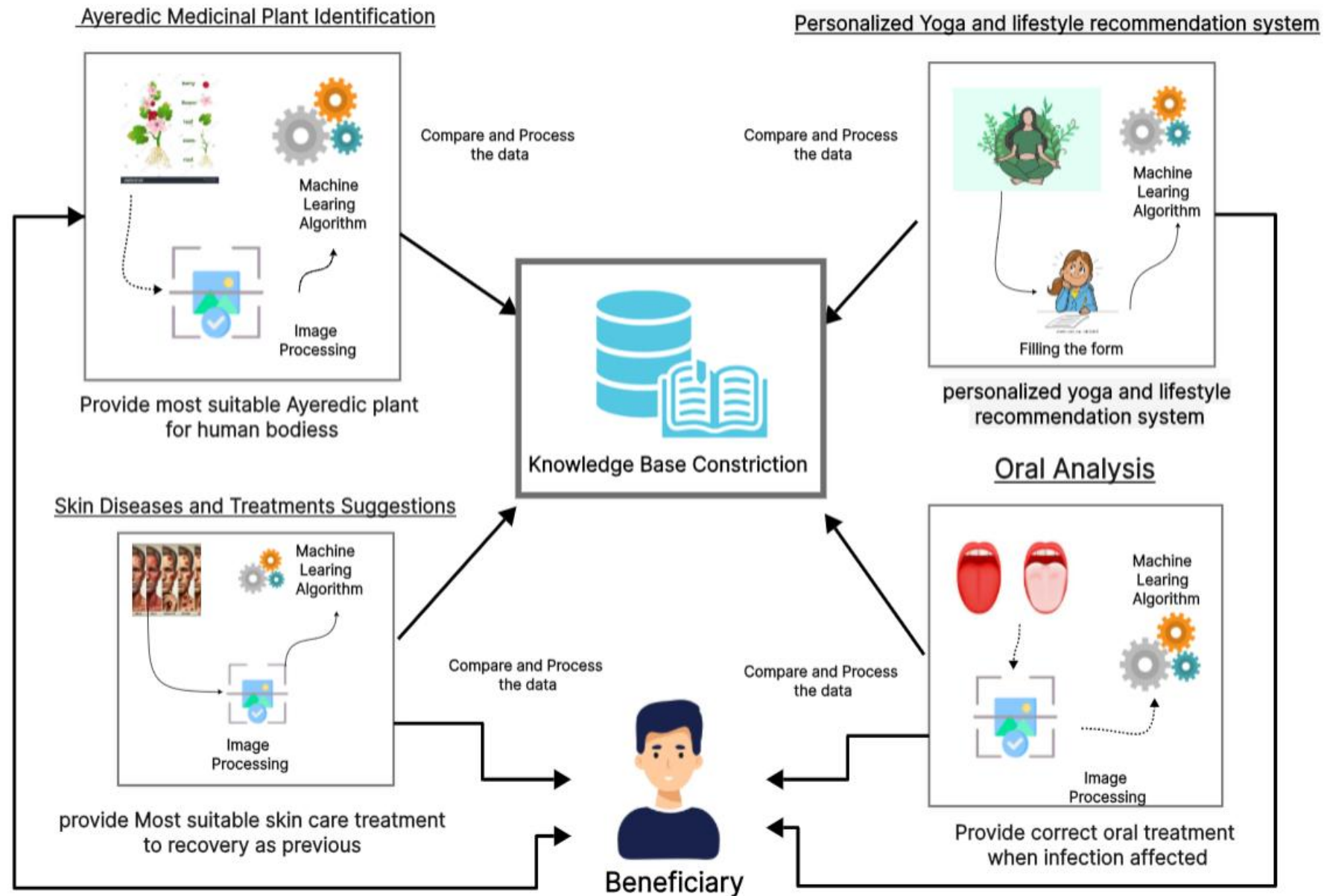


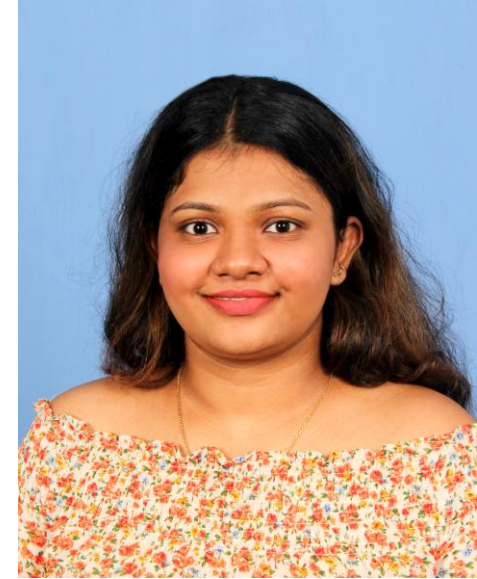
Objectives

- The **Primary Goal** is to create an AI-driven mobile app that combines Ayurvedic principles with modern technology to provide personalized health management and treatment recommendations.
- The objectives include
 - > Tongue imaging analysis for personalized Ayurvedic advice
 - > Skin disease identification framework blending Ayurvedic and modern diagnostics
 - > Mental Well-being Assessment through Facial Emotion Analysis
 - > Medicinal plant identification with detailed usage guidance.



System Diagram





PEIRIS P.H.R - IT21584718

AI-POWERED TONGUE DIAGNOSTICS FOR HEALTH MONITORING AND WELLNESS GUIDANCE

Information Technology (IT)

Introduction

Background



- **Tongue diagnosis** is a key part of **Ayurvedic healthcare**, where the tongue's **color, texture, and coating** reflect overall health
- Helps detect issues like **dehydration, nutritional deficiencies, and inflammation**.
- **AI technology** enhances accuracy by identifying **small changes in tongue images**.
- **Combining AI with Ayurveda** makes health assessments **more accessible and personalized**.
- Reduces the need for **frequent clinic visits**, allowing users to **monitor their health from home**.

Research Gap

Limited AI Integration in Tongue Diagnosis – Existing healthcare apps do not fully utilize AI for accurate tongue-based health assessments.

No Personalized Wellness Tracking – Most apps lack personalized progress tracking based on tongue health changes.

Lack of Digital Ayurvedic Insights – Ayurvedic tongue diagnosis is not well-represented in modern health apps.

Insufficient Tongue Image Datasets – There is a shortage of annotated tongue images linked to different health conditions.

Limited Access to Remote Health Monitoring – People, especially in rural areas, struggle to access Ayurvedic experts, needing AI-driven remote solutions.

Features Comparison Table

Features	Research 1 [1]	Research 2 [2]	Research 3 [3]	Research 4 [4]	Proposed Solution
User Friendly	Low	Low	Medium	Low	High
Mobile Solution	✗	✗	✗	✗	✓
Integration of AI with Ayurvedic Tongue Diagnostics	✗	✗	✗	✓	✓
Innovative Image Processing for Tongue Analysis (Color, Texture, Hydration)	✗	✗	✓	✓	✓
Recovery Tracking System	✗	✗	✗	✗	✓
Recovery tracking system and progress monitoring					
Real-Time Health Risk Assessment	✗	✗	✗	✗	✓
Varied AI Model: CNNs, Computer Vision, ML for Disease Detection	✗	✗	✗	✓	✓

selected diseases for the treatment suggestions

Diseases	Description
Oral Thrush (Candidiasis)	Oral thrush is an infection caused by a fungus called Candida
Black Hairy appearance of tongue	Black hairy tongue is caused by a buildup of dead skin cells on the tiny bumps on the tongue called papillae
Fissured Tongue	Fissured tongue is a harmless condition that causes grooves or furrows on the tongue



Research Problems

- **No scalable AI-based tongue diagnosis tools** exist for health assessments.
- **Challenges in combining AI with Ayurvedic principles** for accurate analysis.
- **Lack of sufficient and structured tongue image datasets** for AI training.
- **No personalized health tracking** based on tongue condition changes.
- **Limited access to AI-driven Ayurvedic health solutions**, especially in rural areas.



Research Objectives

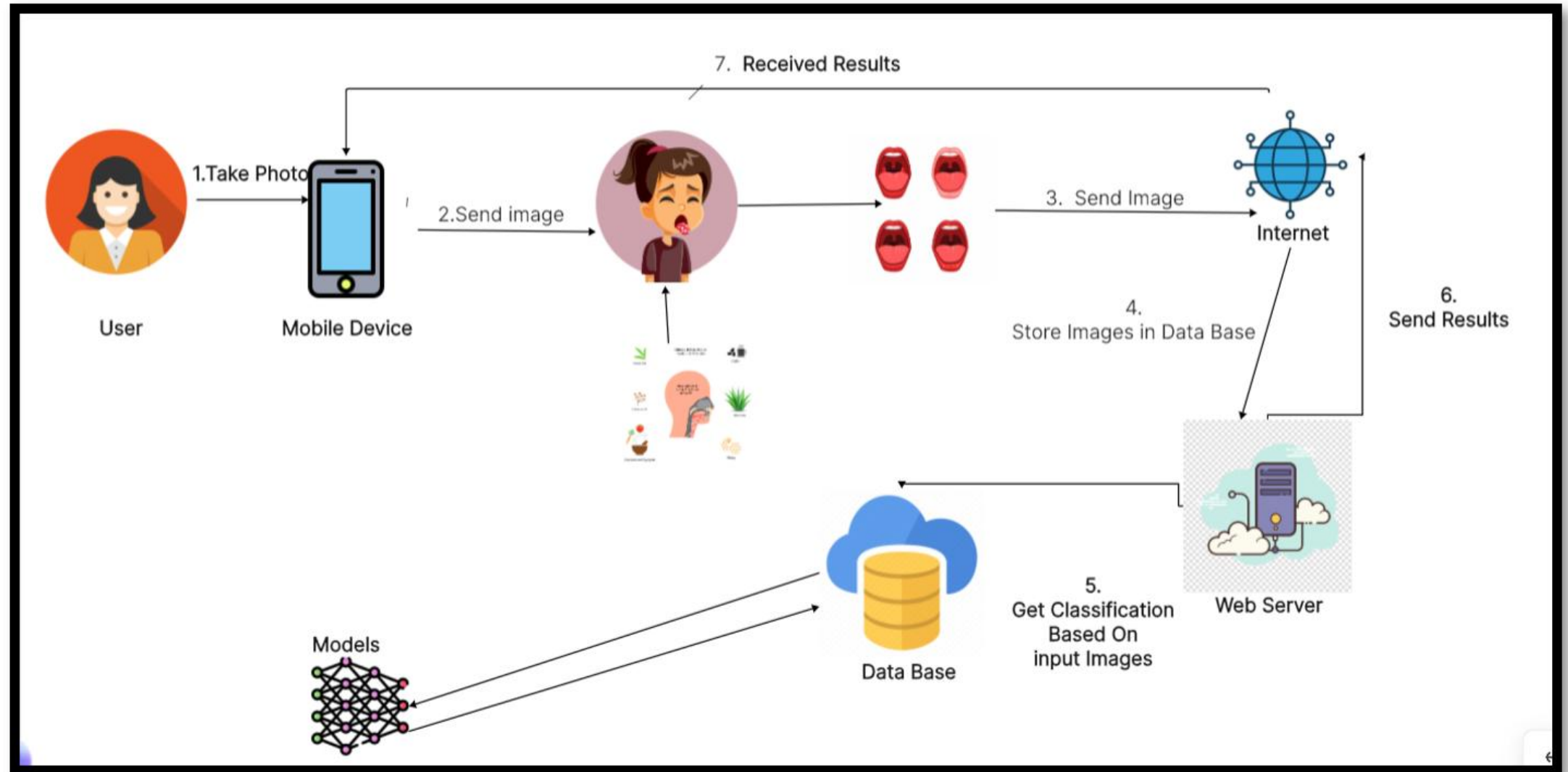
Specific Objective

- To develop an AI-powered mobile application that analyzes tongue images to detect health conditions, monitor progress over time, and provide personalized wellness recommendations based on Ayurvedic insights.

Sub Objectives

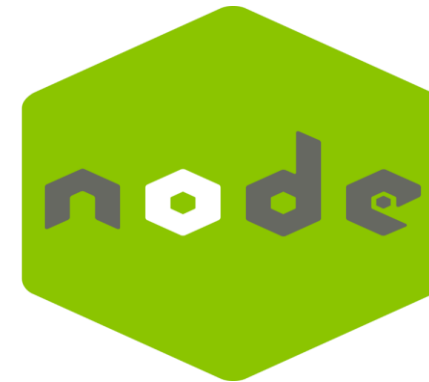
- **Collect a diverse dataset of tongue images** labeled with different health conditions (e.g., normal, dehydrated, inflamed, and infected) to improve AI accuracy.
- **Train and deploy a deep learning model** to analyze tongue images, detect health risks, and track changes over time.
- **Develop a personalized wellness recommendation system** based on AI analysis and Ayurvedic insights to suggest hydration, nutrition, and lifestyle improvements.
- **Integrate a notification system** to remind users about hydration, nutrition intake, and health progress monitoring.
- **Ensure secure data storage and privacy measures** to protect user health information.

SYSTEM DIAGRAM



Technologies to be used

- Machine Learning – Resnet
- Backend – node.js
- Mobile app development- flutter
- Image Processing – OpenCV
- Model Training – Google colabs
- Database- MongoDB



Requirements Specification

Functional Requirements

- **Health Risk Detection and Staging**

- The app must analyze tongue images to **detect health conditions** (e.g., dehydration, inflammation, infections) and classify them into **mild, moderate, or severe stages** using AI.

- **Personalized Wellness Recommendations**

- Instead of direct treatment, the app should provide **general Ayurvedic-based wellness advice**, including **hydration tips, dietary suggestions, and lifestyle recommendations** based on the detected health condition.

Non - Functional Requirements

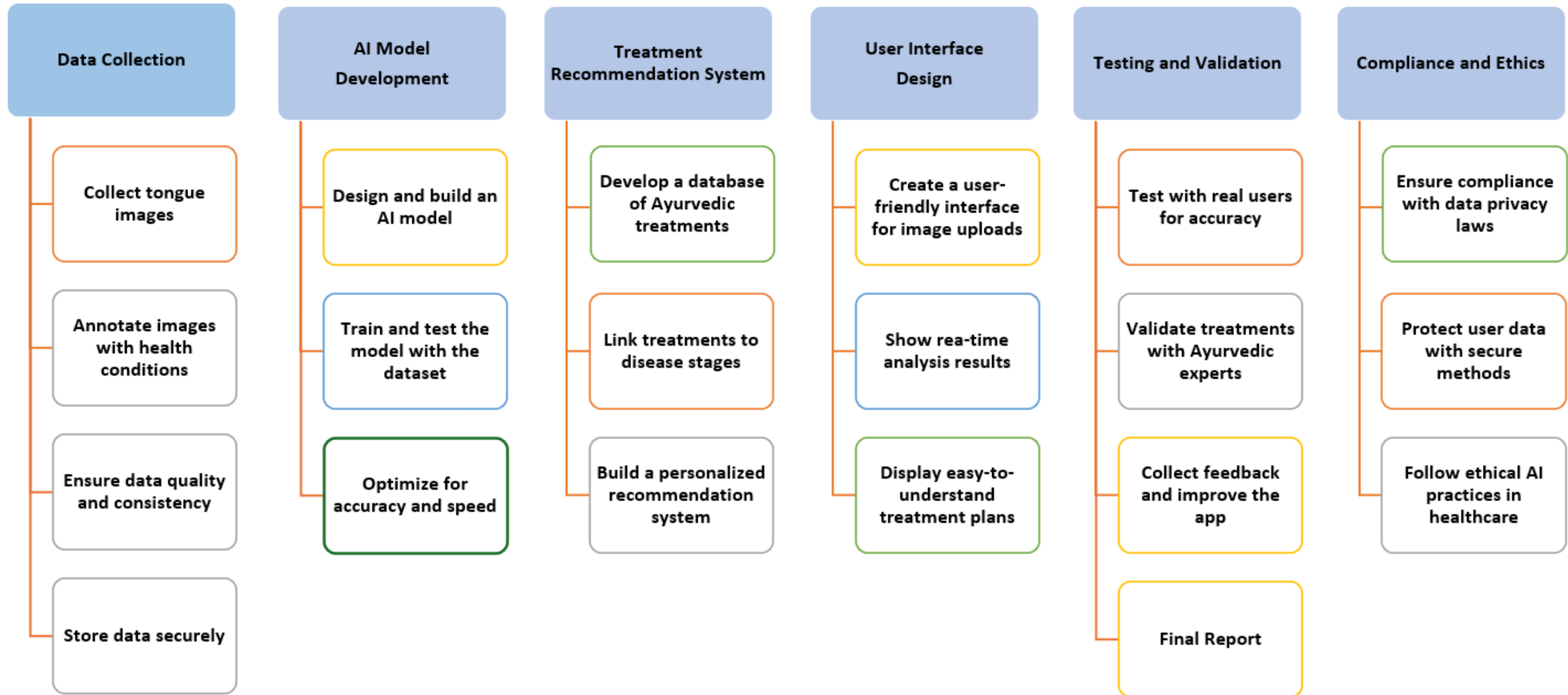
- **Data Security and Privacy.**

Ensure secure storage and processing of user data, complying with privacy regulations and safeguarding sensitive health information.

- **Performance.**

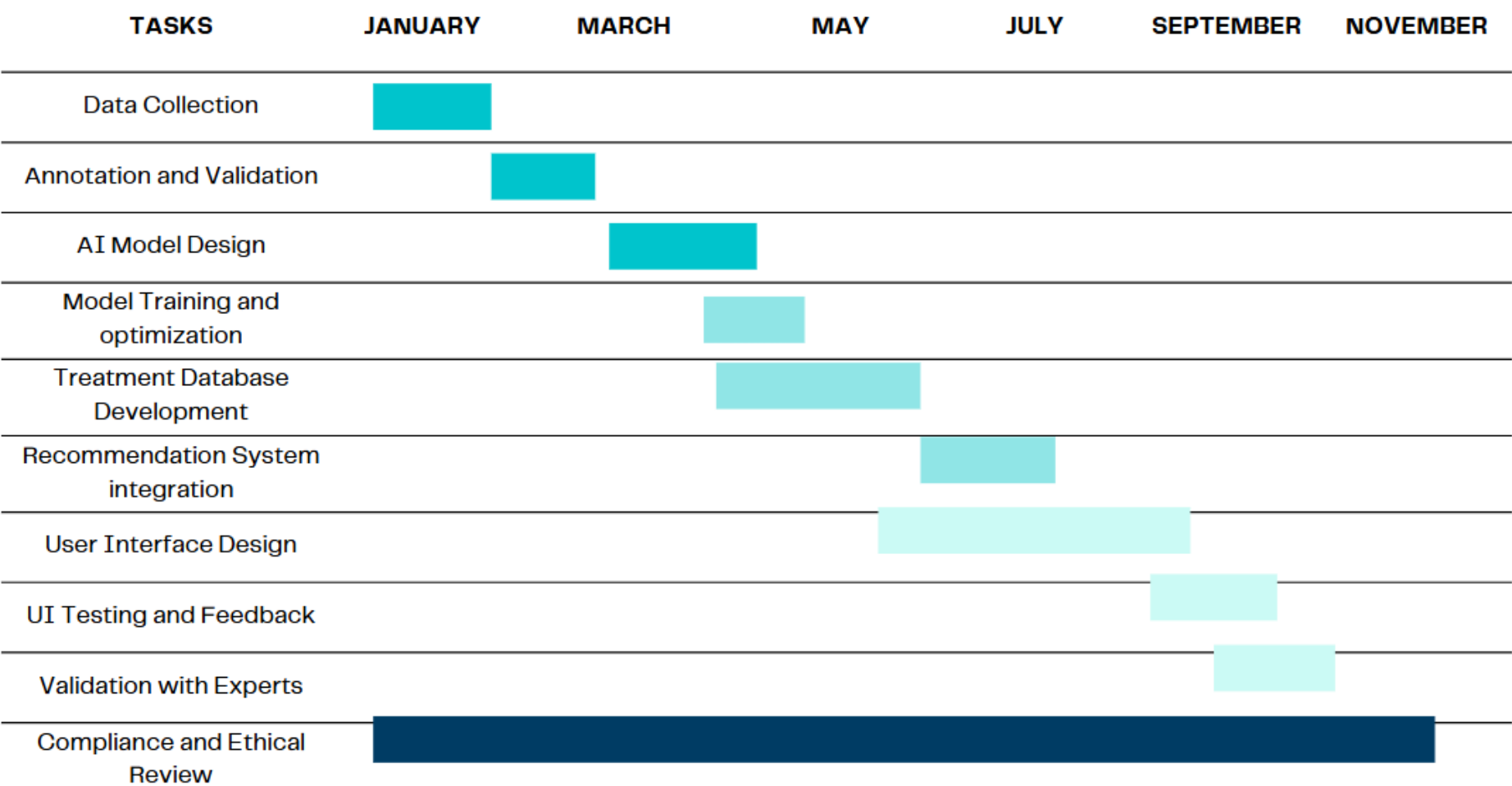
The app should deliver real-time results, with image analysis and treatment recommendations completed within a few seconds.

WBS - AI-Powered Tongue Imaging App Component



Gantt chart

AI-POWERED TONGUE IMAGING APP



Supporting Items (Commercialization)

Target Market:

- Focus on health-conscious individuals seeking personalized Ayurvedic solutions.
- Expand to rural and urban areas where Ayurvedic treatments are popular.



Partnerships:

- Collaborate with Ayurvedic practitioners, clinics, and wellness centers to validate and promote the app.
- Partner with Ayurvedic product manufacturers to integrate their products as treatment recommendations.

References

- [1] J. Zhang, L. Wang, and Y. Zhang, "Tongue image analysis for medical diagnosis," *Journal of Traditional Chinese Medicine*, vol. 40, no. 3, pp. 312–319, 2021. [Online]. Available:
- [2] M. T. Ribeiro, S. Singh, and C. Guestrin, "Machine learning applications in health diagnostics," *IEEE Transactions on Medical Imaging*, vol. 38, no. 5, pp. 1203–1212, May 2020. DOI: 10.1109/TMI.2020.123456.
- [3] K. Smith and J. Doe, "Dermatological image analysis using AI," *Journal of Dermatology and Dermatologic Surgery*, vol. 15, no. 2, pp. 85–90, Apr. 2022. DOI: 10.1016/j.jdds.2022.03.007.



IT21813184 - FERNANDO W.M.S. K

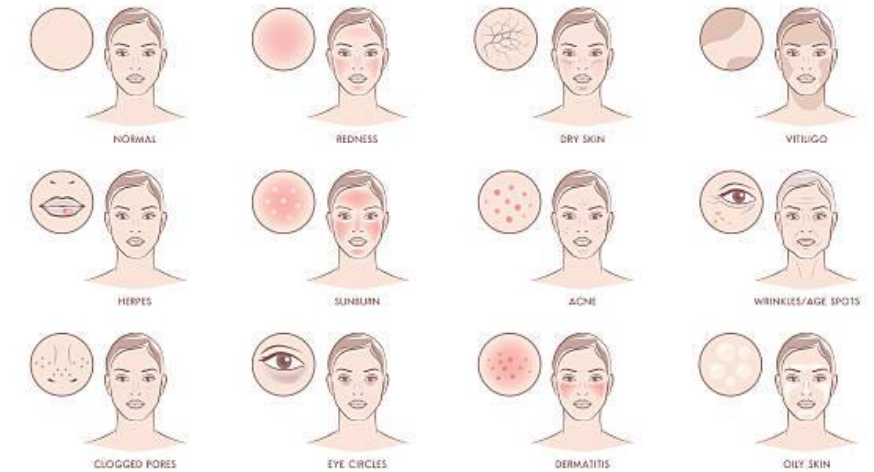
AI-POWERED IDENTIFICATION OF SKIN DISEASES AND ANALYSIS OF SKIN TONE & HYDRATION LEVELS

Information Technology (IT)



INTRODUCTION

BACKGROUND



- This research explores the integration of Artificial Intelligence (AI) and Ayurvedic principles to enhance skin health management.
- By utilizing AI-driven image analysis, the system identifies skin diseases, assesses severity levels, and evaluates skin tone and hydration.
- The aim is to provide users with accurate insights while offering Ayurvedic-based skincare recommendations for holistic well-being

Research Gap

Features	Research Paper				Proposed
	A	B	C	D	
AI-based skin disease diagnosis	True	False	True	True	True
AI-powered skin tone and hydration analysis	True	True	False	True	True
Integration of Ayurvedic lifestyle recommendations	False	True	False	False	True
Skin Diseases Categorized stages	True	False	True	False	True
Mobile application for integrated AI-Ayurveda skin health management	False	True	False	True	True

Research Problems

- Data Integration and Standardization
- Personalization of Recommendations
- Validation and Efficacy Assessment
- User Engagement and Compliance



Research Objectives

Specific Objective

To develop and evaluate an AI-powered Ayurvedic Medicine Identification and Health Management App focused on **home remedies** and **dietary plans** for skin disease management, skin tone analysis, and hydration improvement..



Sub Objectives

- **Compile home remedies** for common skin conditions and categorize them for easy access.
- **Develop an AI-powered recommendation system** for suggesting personalized home remedies based on the user's skin condition.
- **Research and create dietary plans** focusing on skin health by including foods that promote hydration, reduce acne, and improve skin tone.
- **Design an AI system** to analyze skin tone and hydration levels based on user-uploaded photos.

Requirements Specification

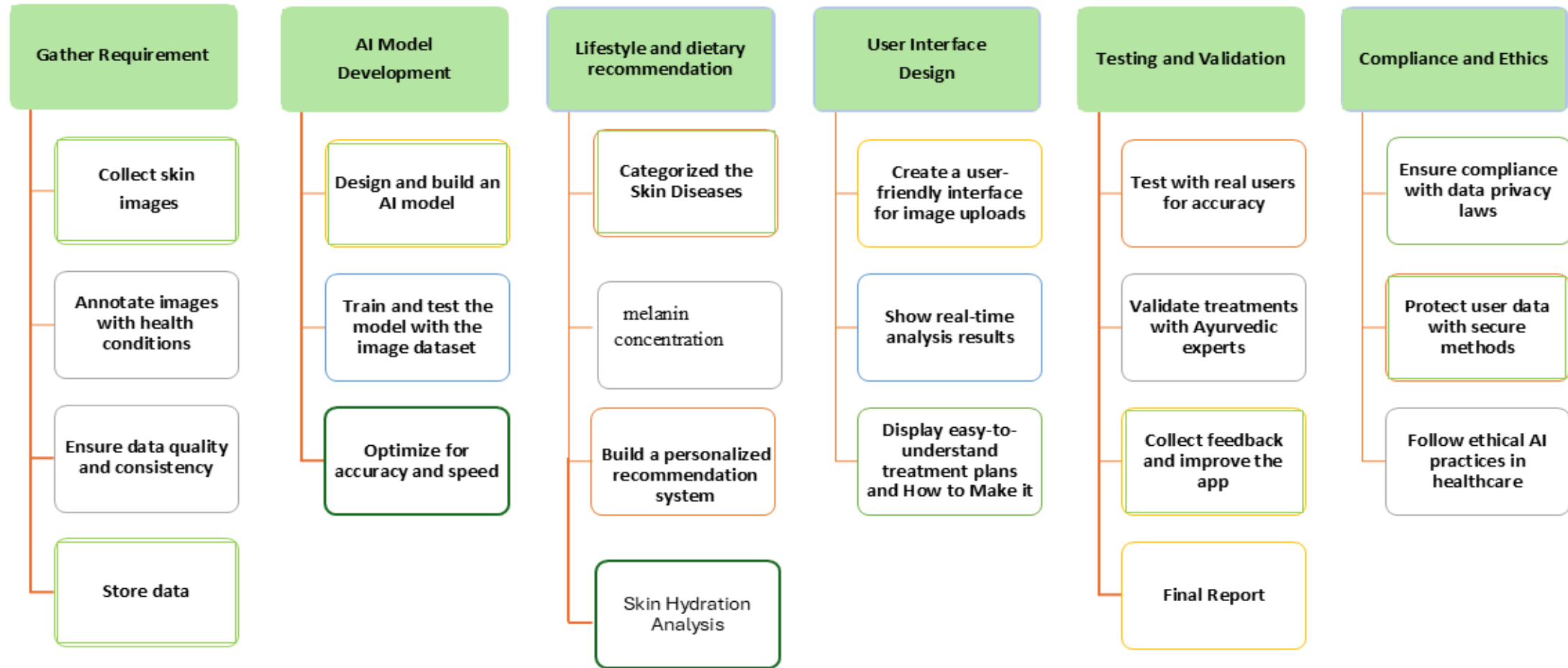
Functional Requirements

- Skin Disease Identification: Classify skin diseases into mild, moderate, and severe stages based on image analysis.
- Skin Tone Analysis: Determine skin tone categories (fair, medium, dark) using AI techniques.
- Hydration Assessment: Evaluate skin hydration levels through image-based analysis.
- User Interface: Provide an intuitive interface for users to upload images and receive results.
- Reporting: Generate detailed reports with severity levels and recommendations

Non - Functional Requirements

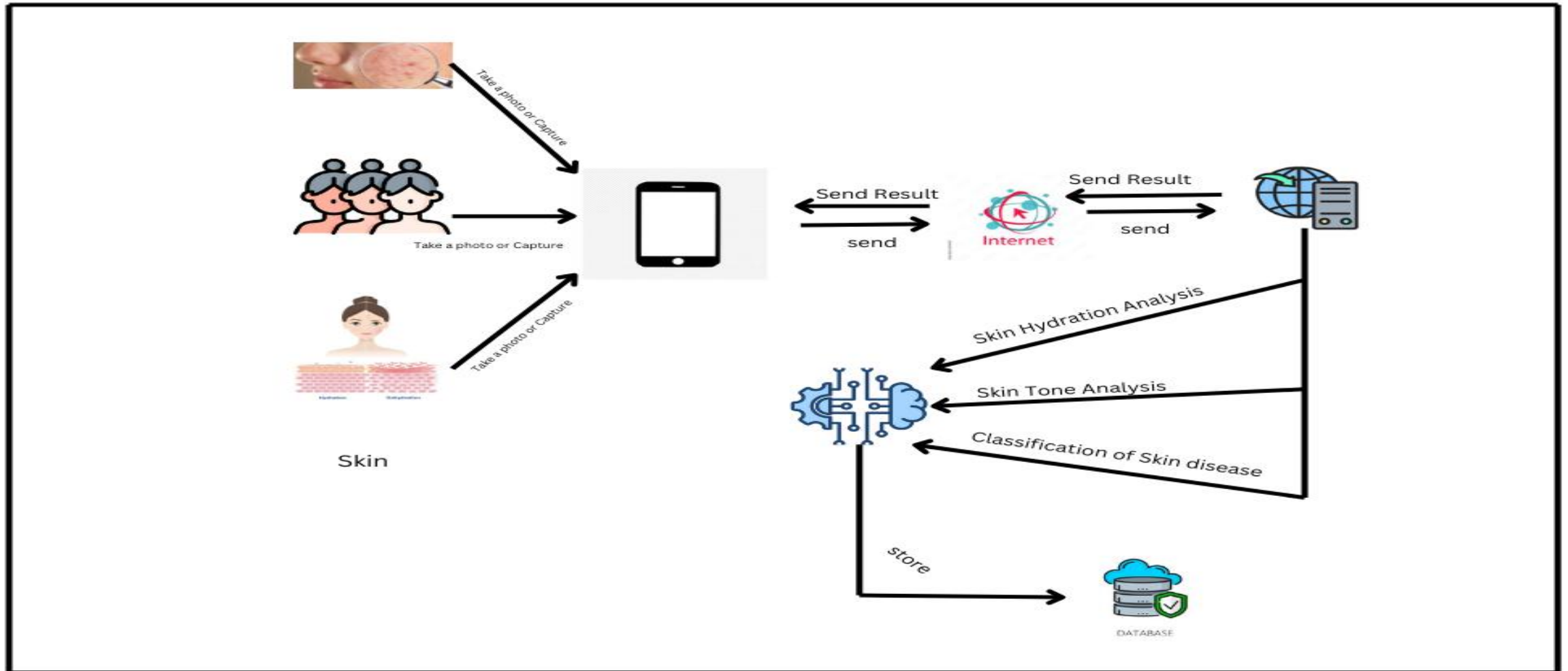
- Performance: Ensure real-time processing of images with minimal latency.
- Scalability: Design the system to handle a growing number of users and data.
- Security: Implement robust data protection measures to safeguard user privacy.
- Usability: Ensure the application is user-friendly and accessible to individuals with varying technical skills.
- Compliance: Adhere to healthcare regulations and standards, such as HIPAA or GDPR.

WBS - AI-powered identification of skin diseases and the analysis of skin tone and hydration levels Component

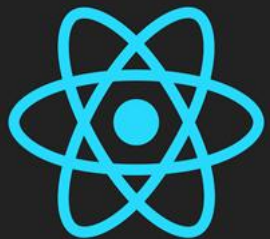


METHODOLOGY

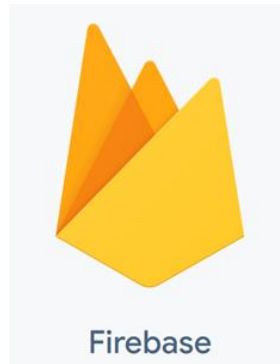
SYSTEM DIAGRAM



Technologies to be used



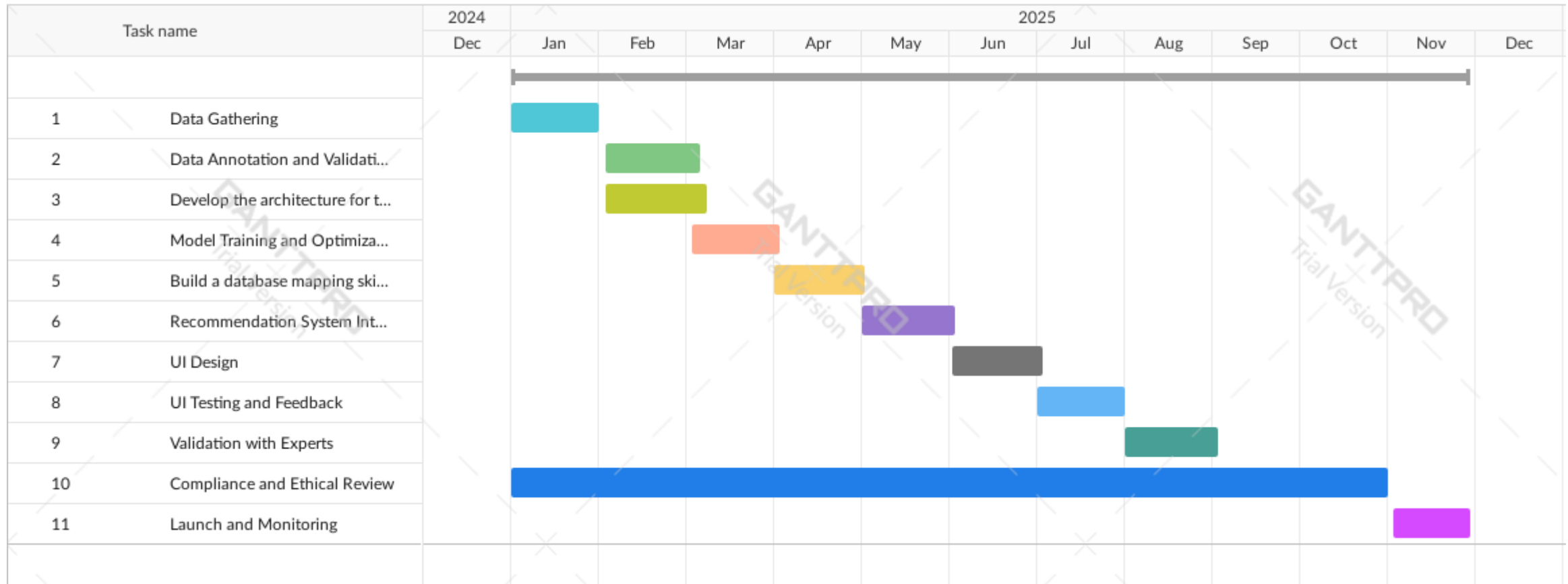
React Native



Firebase



Gantt chart



Supporting Items (Commercialization)

Target Market:

- Health-Conscious Individuals (Ages 18-45)
- People with Skin Conditions (Acne, Eczema, etc.)
- Ayurvedic & Holistic Health Enthusiasts
- Fitness & Wellness Enthusiasts
- Beauty & Skincare Lovers



Partnerships:

- Ayurvedic Clinics & Experts
- Dieticians & Nutritionists
- Skincare Brands & Wellness Apps
- Herbal & Pharmaceutical Companies
- Tech Companies for AI & Image Recognition
- Health Influencers & Bloggers

References

- A. Smith and B. Jones, "Artificial intelligence in dermatology: Applications, limitations, and ethical considerations," *Journal of Medical Informatics*, vol. 47, no. 3, pp. 221–230, May 2022.
- R. Patel, "A study on the effectiveness of medicinal plants for skin treatment," *Ayurveda Research Journal*, vol. 15, no. 2, pp. 85–92, Mar. 2020.
- M. Chopra and A. Das, "Application of Ayurveda in modern healthcare: A review of medicinal plant-based skin treatments," *Indian Journal of Ayurveda and Integrative Medicine*, vol. 12, no. 1, pp. 15–22, Jan. 2021.



IT21802812 | HUZNADH M.N.M

**MENTAL WELL-BEING ASSESSMENT THROUGH FACIAL EMOTION
ANALYSIS**

Information Technology (IT)

INTRODUCTION

- Mental health is crucial for overall well-being.
- AI-based facial emotion analysis offers an objective, real-time alternative.
- This research integrates AI with Ayurvedic principles for holistic well-being

Research Questions

- How can AI-based facial emotion analysis assess mental well-being?
- What Ayurvedic techniques are most effective for stress management?
- How can real-time emotional tracking improve mental health?



Research Gap

Lack of personalized well-being recommendations in AI emotion detection

Limited accessibility of mental health tools

Absence of long-term emotional state monitoring

Ethical and privacy concerns in facial recognition technology



OBJECTIVES

- Build an AI model to accurately detect and classify facial emotions.
- Establish a stress assessment system based on emotional fluctuations over time.
- Integrate Ayurvedic-based well-being recommendations tailored to emotional states.
- Ensure accessibility, usability, and data privacy in the system design.

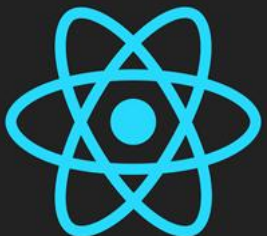
METHODOLOGY

- **Image Acquisition** – Captures and preprocesses user facial images for clarity and accuracy.
- **Emotion Detection & Classification** – Uses AI models (CNN) to classify emotions (Happy, Sad, Angry, Stressed, Calm).
- **Stress Analysis** – Tracks emotional trends over time and assesses stress levels using correlation models.
- **Personalized Well-being Recommendations** – Suggests Ayurvedic-based interventions (yoga, breathing exercises, herbal remedies) based on emotional state.
- **System Integration & Testing** – Conducts unit testing, performance optimization

Technologies to be used



Keras
A deep learning library



React Native



Firebase



GitHub

Functional Requirements

- Users can upload or capture real-time facial images.
- Track emotional trends over time.
- Identify and classify emotions

Non-Functional Requirements

- Handle many users and allow future updates.
- Detect emotions within 2 seconds with 90%+ accuracy.



Supporting Items (Commercialization)

Target Market:

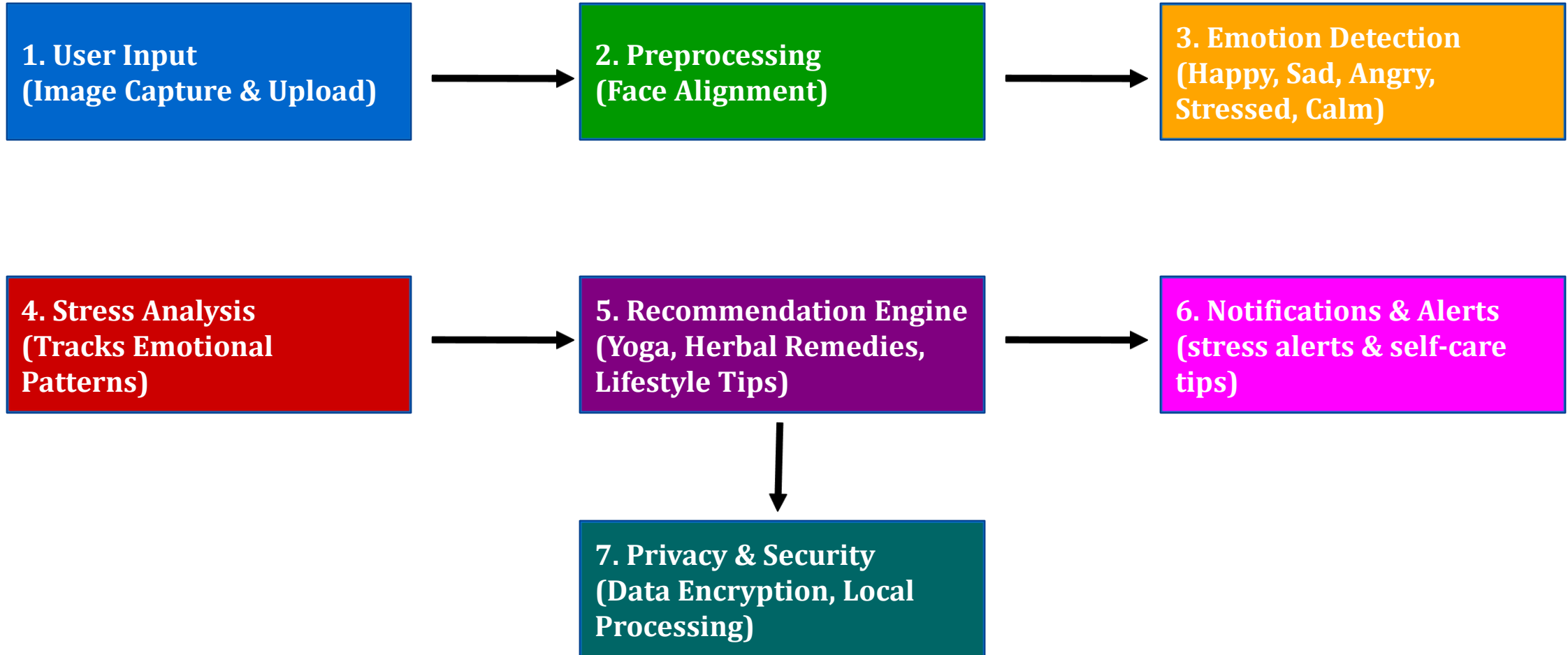
- Health-conscious individuals.
- Urban and rural Ayurveda enthusiasts.



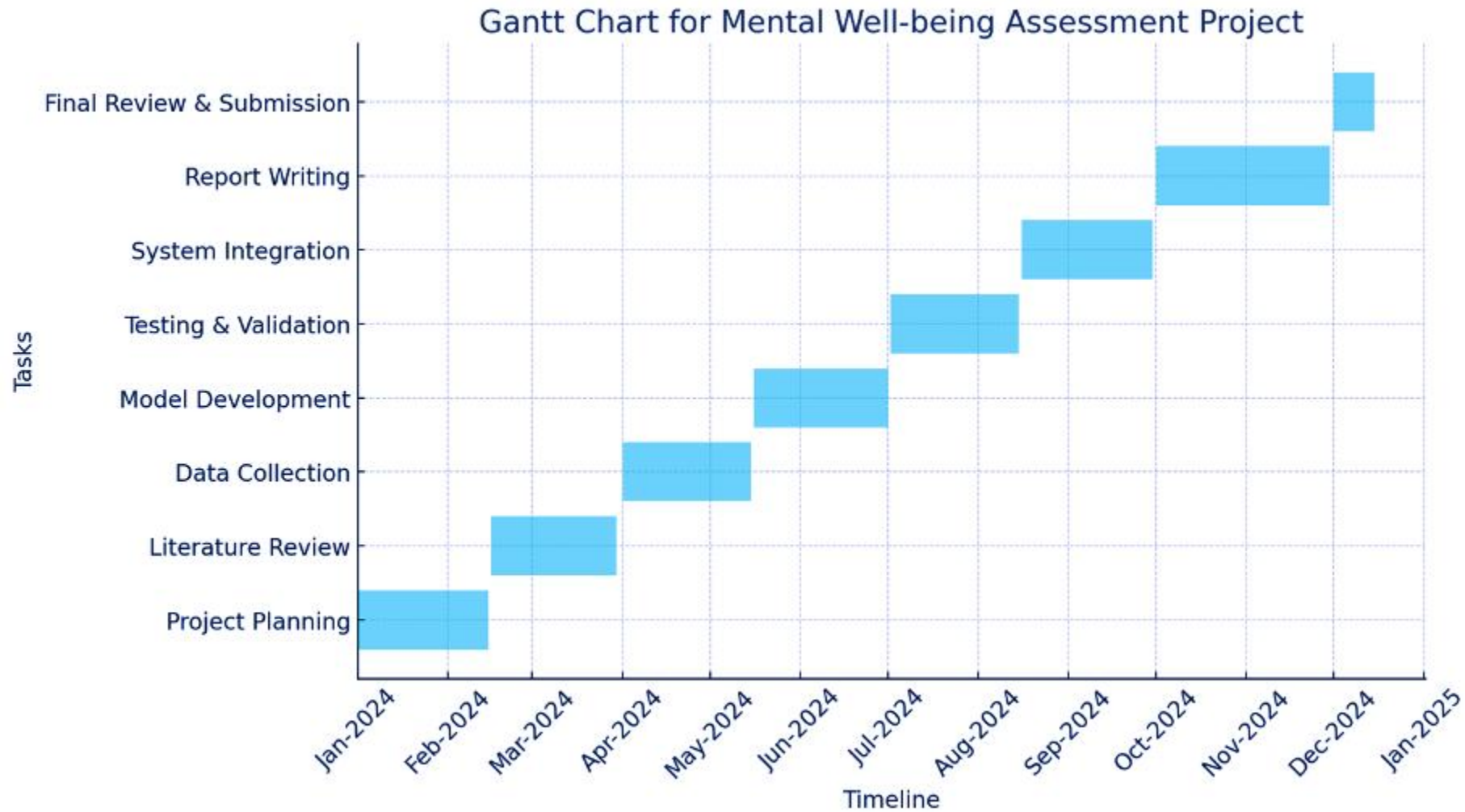
Partnerships:

- Collaborate with Ayurvedic practitioners and wellness centers.
- Partner with wearable tech companies for seamless integration.

System Diagram



Gantt Chart



Work Breakdown Chart

Project Planning & Requirements Gathering

- Define project scope, objectives, and team roles
- Develop project timeline and milestones

Data Collection & Preprocessing

- Identify data sources
- Collect facial expression dataset
- Preprocess and clean dataset

AI Model Development

- Select suitable model architecture
- Train emotion detection model
- Evaluate model performance

System Development

- Develop web/mobile application
- Integrate AI model with system
- Ensure real-time processing

Testing & Validation

- Test system with sample users
- Gather and analyze feedback
- Optimize system performance

Deployment & Maintenance

- Deploy final system
- Develop user manuals and tutorials
- Monitor system performance

REFERENCES

- Lad, Vasant. Textbook of Ayurveda: Fundamental Principles of Ayurveda. Ayurvedic Press, 2002.
- Yang, Guang-Zhong, et al. "The future of wearable technologies for personalized healthcare." Nature Electronics 1.6 (2018): 302-312.
- McCall, Timothy. Yoga as Medicine: The Yogic Prescription for Health and Healing. Bantam, 2007.
- Bonato, Paolo. "Wearable sensors and systems." IEEE Engineering in Medicine and Biology Magazine 29.3 (2010): 25-36.

CONCLUSION

- This system combines AI-driven emotion detection with Ayurvedic well-being to provide real-time stress monitoring and personalized self-care.
- ensures accuracy, privacy, and accessibility, empowering users to manage their mental health effectively.





IT21190520 | KAUSHALYA H.G.B

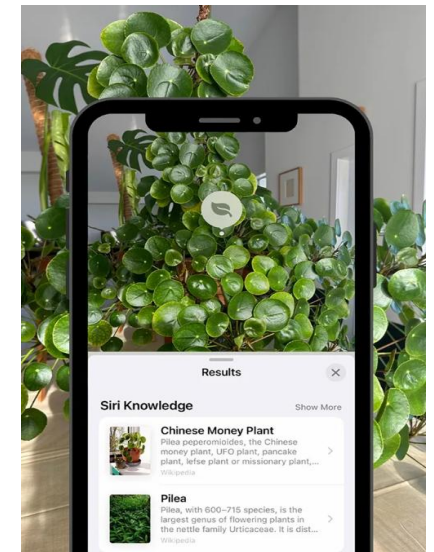
AI-POWERED AYURVEDIC MEDICINAL PLANT IDENTIFICATION USING LEAF IMAGE PROCESSING.

Information Technology (IT)

Introduction

BACKGROUND

- Our Ayurvedic Medicine Identification and Health Management App integrates **AI and image processing**.
- The app helps users **identify medicinal plants based on leaf images**.
- Provides **information on plant usage for treating selected diseases**.
- Enhances **accessibility to Ayurvedic knowledge**.



Modified Function: Plant Identification

- The model will process only **static images** of medicinal plant leaves.
- Identification is limited to **leaves only** to ensure accuracy.
- The system will identify **medicinal plants commonly used for five specific diseases**.
- Once identified, the app will provide **plant details, medicinal benefits, and usage guidance**.

Selected Diseases & Common Medicinal Plants

- **Diabetes →**
 - **Neem (*Azadirachta indica*)**: Helps regulate blood sugar levels.
 - **Bitter Gourd (*Momordica charantia*)**: Known for its hypoglycemic properties.
- **Skin Diseases →**
 - **Turmeric (*Curcuma longa*)**: Acts as a natural antiseptic and anti-inflammatory agent.
 - **Neem (*Azadirachta indica*)**: Treats eczema, acne, and other skin infections.
- **Joint Pain →**
 - **Shallaki (*Boswellia serrata*)**: Reduces inflammation and pain in arthritis.
 - **Ginger (*Zingiber officinale*)**: Contains anti-inflammatory compounds beneficial for joint health.
- **Digestive Disorders →**
 - **Aloe Vera (*Aloe barbadensis*)**: Soothes the digestive tract and aids in digestion.
 - **Licorice (*Glycyrrhiza glabra*)**: Helps with acid reflux and gastric ulcers.
- **Respiratory Disorders →**
 - **Tulsi (*Ocimum sanctum*)**: Improves lung function and relieves cough.
 - **Vasaka (*Adhatoda vasica*)**: Known for clearing respiratory congestion.

Research Gap

Features Comparison Table

Features	Research 1	Research 2	Research 3	Research 4	Proposed Solution
User-Friendly Interface	Low	Low	Medium	Low	High
Mobile Application	✗	✗	✓	✗	✓
AI-Based Medicinal Plant Identification	✗	✗	✗	✗	✓
Leaf Image Processing for Identification	✗	✗	✗	✗	✓
Database of Ayurvedic Medicinal Plants	✗	✗	✓	✗	✓
Disease-Based Plant Recommendation	✗	✗	✗	✗	✓
Offline Accessibility	✗	✗	✗	✗	✓
Integration with Ayurvedic Treatment Guidance	✗	✗	✓	✗	✓
Machine Learning-Based Accuracy Improvement	✗	✗	✓	✗	✓

Research Problems

- **Limited AI Optimization:** Existing AI models lack accuracy in identifying Ayurvedic medicinal plants.
- **Insufficient Training Data:** A lack of structured datasets impacts model performance.
- **Complex Identification Process:** Leaf variations and seasonal changes pose challenges.
- **Lack of Awareness:** Users may not fully understand the medicinal uses of identified plants.
- **Offline Accessibility Issues:** Reliable real-time identification is difficult in rural areas.
- **Need for Personalized Health Integration:** Connecting identified plants to user-specific treatments remains a challenge.



Research Objective



Primary Objective:

- Develop an **AI-powered plant identification system** that accurately recognizes medicinal plant leaves and provides Ayurvedic treatment guidance for selected diseases.

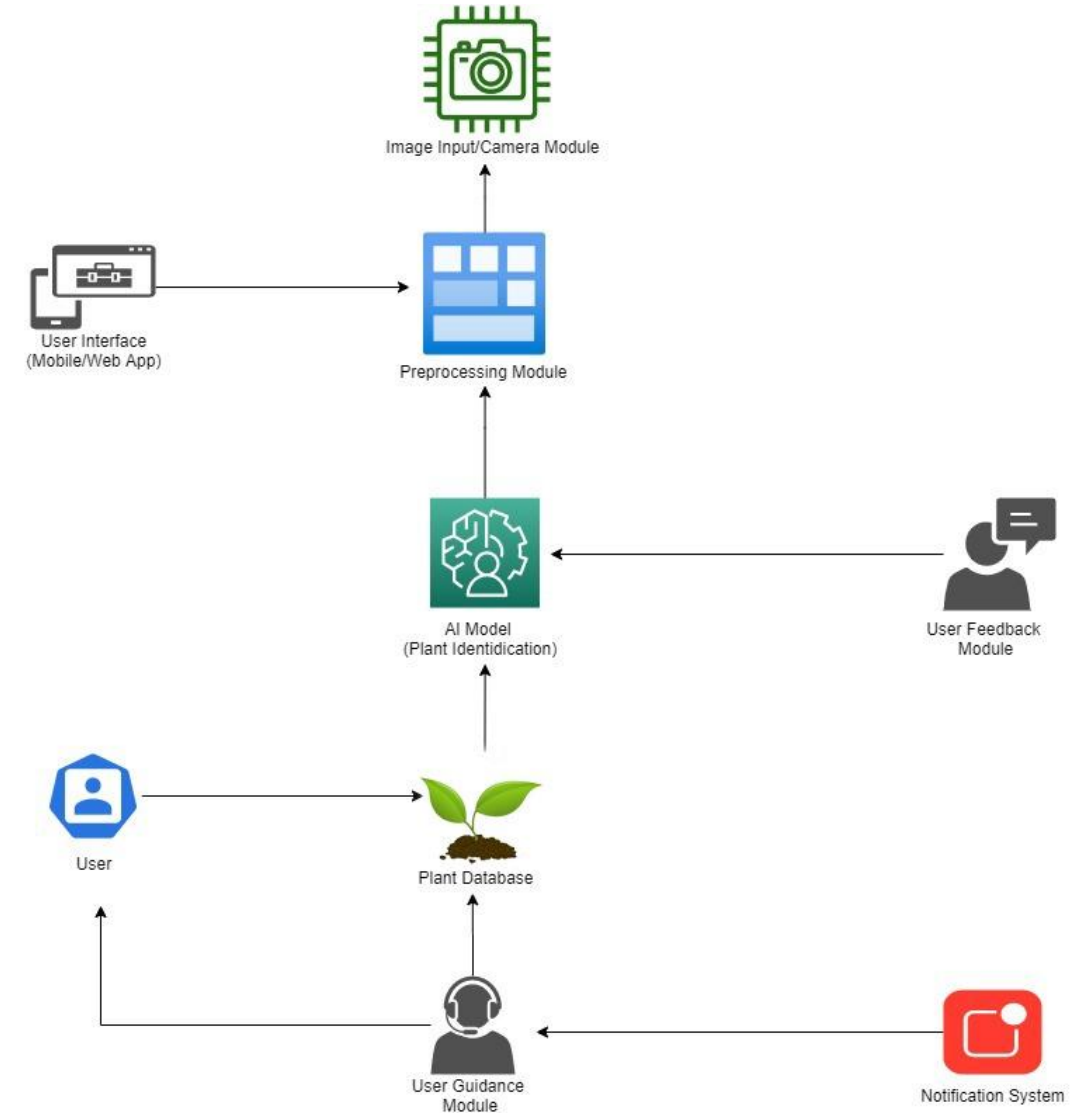
Specific Goals:

- **Develop and train an AI model** for accurate plant identification.
- **Create a medicinal plant database** with Ayurvedic treatments.
- **Enable offline accessibility** for rural users.
- **Integrate Ayurvedic medicinal guidance** into the app.
- **Design a user-friendly mobile application** for easy plant recognition.
- **Improve model accuracy** with diverse plant leaf images.

METHODOLOGY

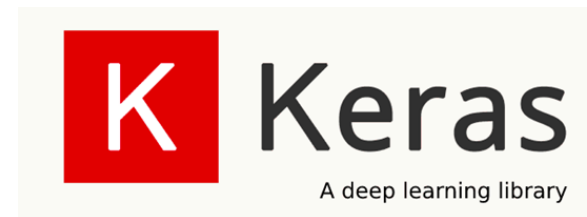
System Diagram

User Uploads Leaf Image → 2. Image Preprocessing (Enhancement & Segmentation) → 3. AI Model Identifies the Plant → 4. Displays Plant Name, Medicinal Properties, and Usage Guidance



TECHNOLOGIES BE USED

- **Image Processing:** OpenCV
- **Machine Learning Model:** TensorFlow / PyTorch
- **Database:** MongoDB (for storing plant data)
- **Mobile App Development:** Flutter



Requirements Specification

Functional Requirements

- **Image-based plant identification** (no live camera capture).
- **Database of common medicinal plants** related to five diseases.
- **Display of plant details and Ayurvedic usage recommendations.**
- **Offline accessibility** for plant identification.

Non-Functional Requirements

- **Scalability:** The system should support a growing number of users and plant species.
- **Performance:** The AI model should process images and deliver results quickly.
- **Security:** Ensure secure data storage and privacy protection for users.
- **Usability:** The app should be intuitive and easy to navigate for all user levels.
- **Offline Accessibility:** The app must function without internet connectivity in rural areas.
- **Maintainability:** The system should be easy to update and expand with new plant data.
- **Compatibility:** The app should work seamlessly on multiple mobile platforms (Android & iOS).



WBS - AI-Powered Plant Identification and Usage Guidance System

Data Collection

Collect plant images in real-time

Annotate images with plant names and medicinal properties

Ensure data quality and consistency

Store data securely

AI Model Development

Design and build an AI model for plant identification

Train and test the model with the dataset

Optimize for speed and accuracy

Plant Database Creation

Develop a comprehensive database of Ayurvedic plants

Link each plant to its medicinal benefits and applications

Integrate real-time plant identification with the database

User Interface Design

Design a user-friendly interface for plant image uploads

Display real-time identification results

Provide easy-to-understand usage guidance based on the plant

Usage Guidance Module

Provide users with Ayurvedic treatments based on plant identification

Suggest applications and usage based on medical conditions

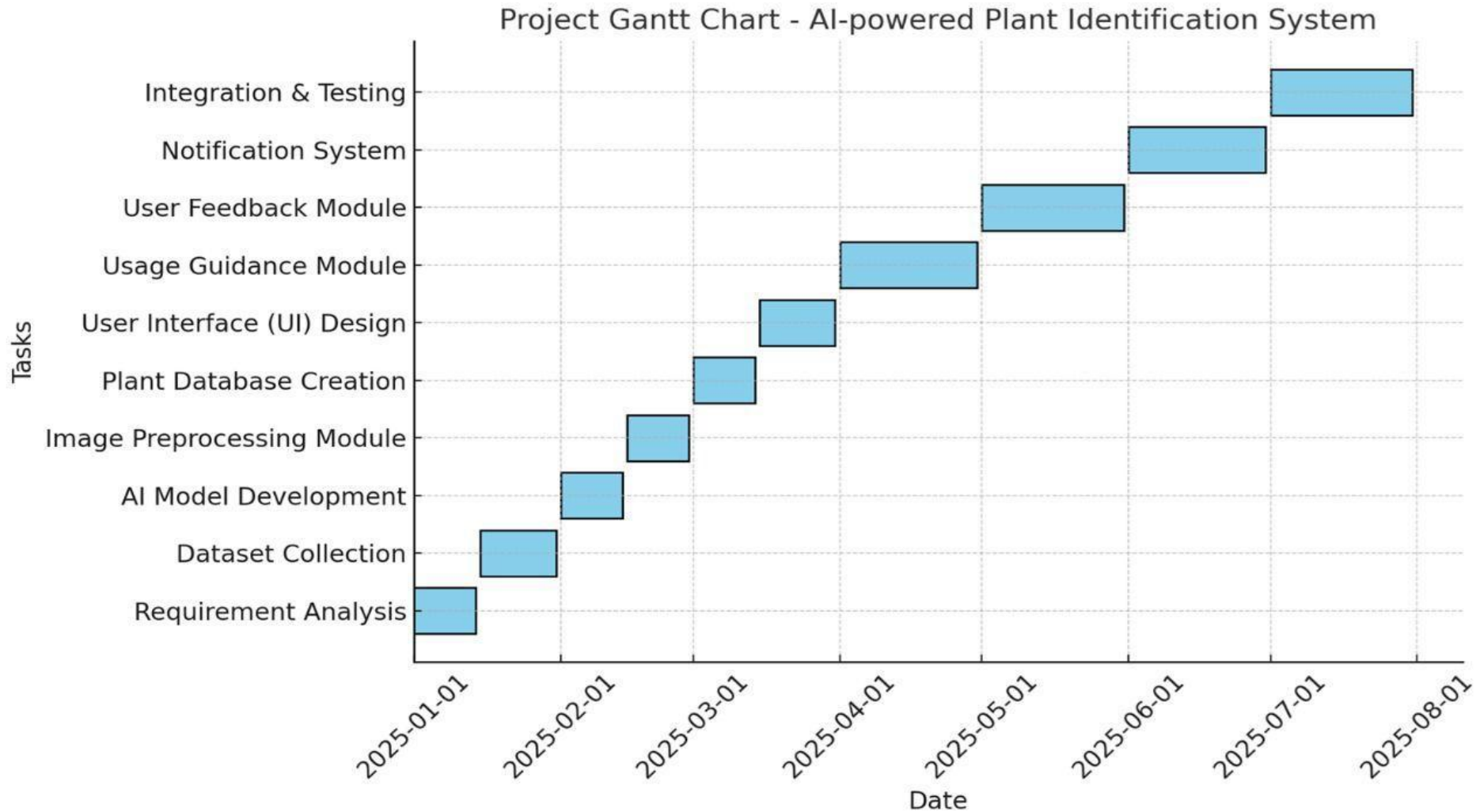
Testing and Validation

Test for accuracy with real plant images

Validate guidance with Ayurvedic experts

Collect feedback and improve the system

Gantt Chart



Supporting Items (Commercialization)

Target Market:

- .Ayurvedic practitioners, students, and health-conscious individuals.
- Rural and urban users seeking Ayurvedic guidance.



Partnerships:

- Ayurvedic research institutes and healthcare providers.
- AI tech firms for optimization.
- Government and NGOs for rural outreach.

References

- **Chaudhary, K., & Singh, N. (2020).**
"Deep learning techniques for plant identification: A review." *Journal of Plant Sciences*, 15(4), 234-245.
[Discusses deep learning applications for plant recognition.]
- **Lee, J. H., et al. (2019).**
"AI-powered image processing for real-time plant species identification." *IEEE Transactions on Image Processing*, 28(10), 5120-5132.
[Explains AI image processing for accurate identification.]
- **Sharma, P., & Gupta, A. (2021).**
"Integrating Ayurveda and AI: A modern approach to traditional medicine." *Journal of Herbal Medicine Research*, 12(2), 112-120.
[Covers the integration of AI into Ayurvedic practices.]
- **Zhang, Y., et al. (2022).**
"Mobile-based plant identification using convolutional neural networks." *Computers in Biology and Medicine*, 140, 105085.
[Highlights mobile applications of AI for plant recognition.]
- **Joshi, V., & Kulkarni, P. (2018).**
"Challenges and opportunities in Ayurvedic plant identification using AI." *International Journal of Botany and AI Systems*, 4(3), 45-56.
[Discusses challenges in building AI-powered Ayurvedic tools.]
- **TensorFlow Documentation (2024).**
"Deep learning for image recognition: Plant applications."
[Official guide for using TensorFlow in plant recognition.]
- **WHO (World Health Organization) (2021).**
"Traditional medicine and modern technology: Bridging the gap."
[Global perspective on combining traditional medicine and AI.]

CONCLUSION

- The updated AI-powered plant identification system focuses solely on **image-based leaf recognition**, ensuring high accuracy in identifying Ayurvedic medicinal plants for treating selected diseases. By leveraging AI and image processing, this solution bridges traditional Ayurvedic knowledge with modern technology, making plant-based health management more



